

Data Viz Draft

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Pre-planning

Restate question hoping to answer in infographic

1. overarching question (overall theme) What citizen science reveals about California Bats.
2. three subquestions (each will addressed in component visualizations)

My questions have changed since FPM #1 because before I didnt have data but once I found data I had to modified what I could do based on the information the data provides. Updated questions are as follow;

- What species are out there?
 - Where are bats found?
 - When are they most active?
2. The following variables will be used to answer the above questions.

My data comes from the Calbat project in iNaturalist with 1843 citizen science observation.

To answer what species are out there the following variables will be used: - scientific_name; species identification - common_name; for general audience labels instead of using scientific names. I will need to wrangle the data and relabel for clarity - quality_grade; to show research vs needs_id (citizen science angle) - num_identification_agreements; shows community verification process

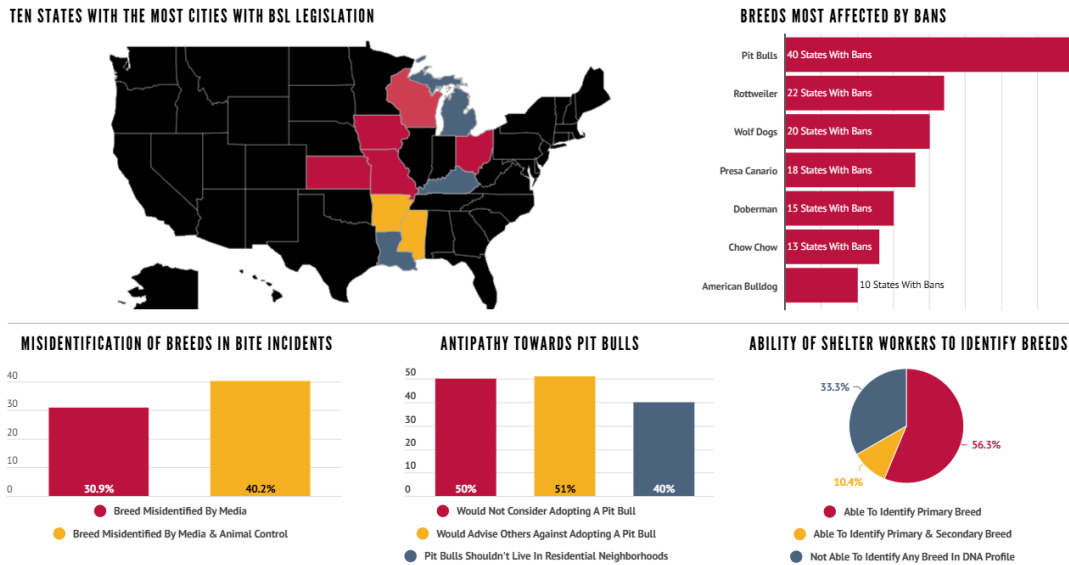
To answer where bats are found: - latitude and longitude; geographic coordinates will need to convert to spatial objects - California boundaries will need to query from **tigris** package - place_guess; location context

To answer when bats are most active: - observed_on; observation dates Will need to extract month for seasonal pattern and year for filtering

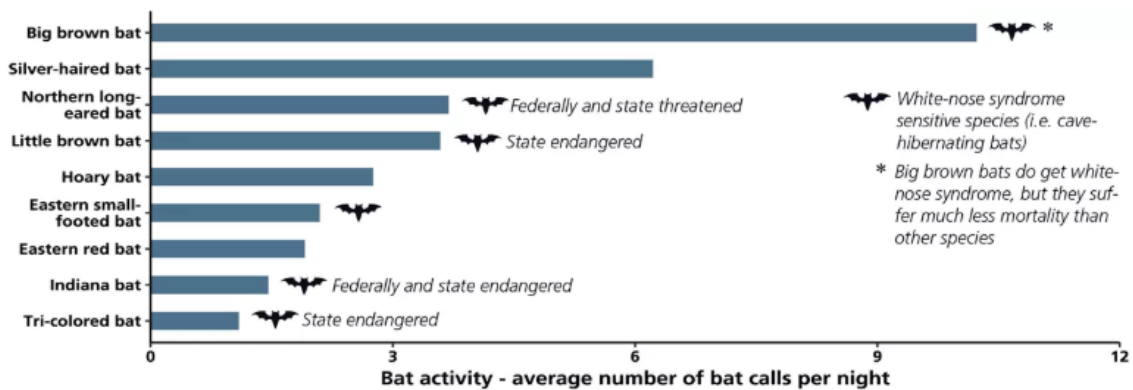
May also need: - user_login; to count unique observers - id; total observation count

3. The following images have elements that I will borrow for my visualizations

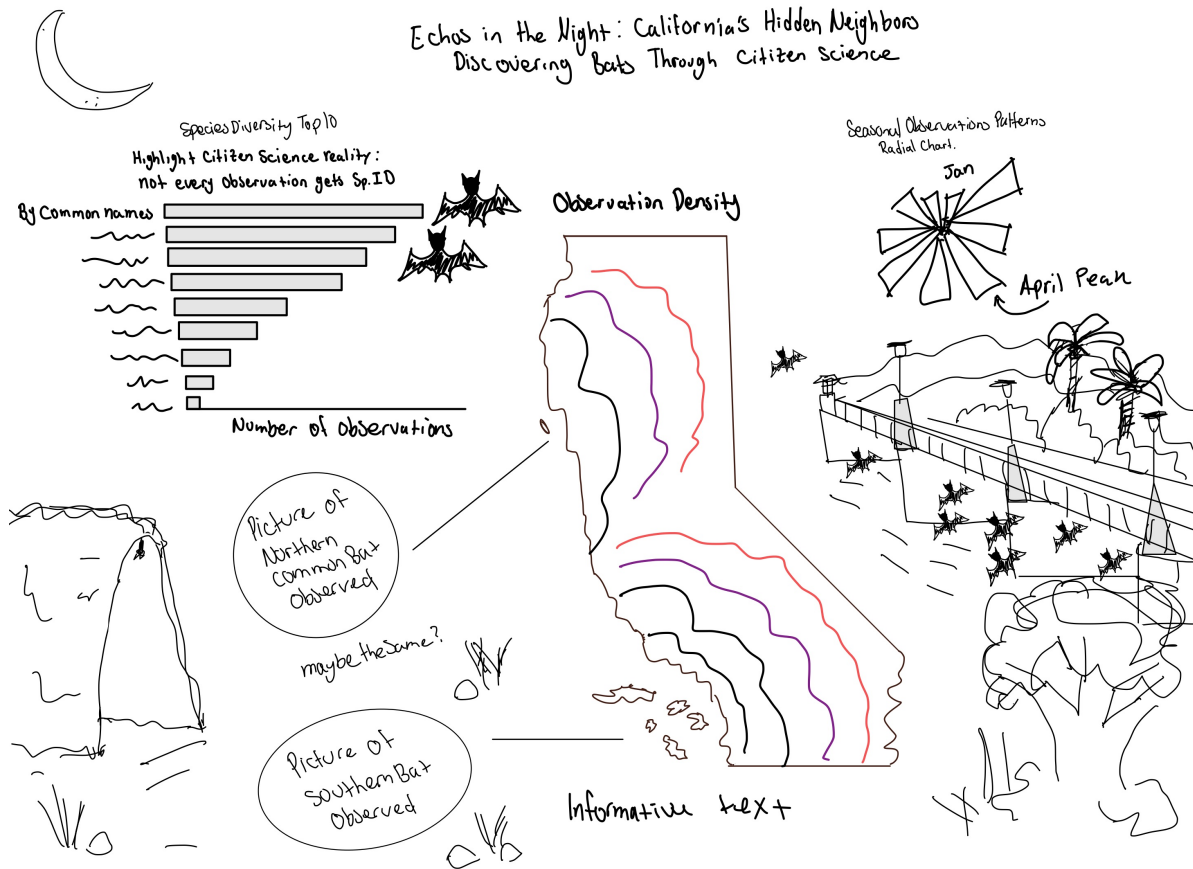
The figure below has a nice layout. I also plan to make a map plot. I feel like it will take a lot of space and this gives me an idea on how I can arrange my plots. I also like the bar plots which have a lot of annotations on them but are not disturbing. Gives me some idea on how I can add information without being overwhelming.



On the plot below I like how they used bat symbols to indicate which bats are sensitive to white-nose syndrome, used like some type of legend design. It can be something I can use on my species plot.



IV. Hand-drawn anticipated visualization



V. Recreated hand-drawn visualization using code

```
#.....load libraries.....
library(tidyverse)
library(here)
library(lubridate)
library(sf)
library(tigris)
options(tigris_use_cache = TRUE)
#.....import data.....

calbats <- read_csv(here("data", "observations-675568.csv"))
```

```

# clean data 1018

bats_clean <- calbats %>%
  filter(
    captive_cultivated == FALSE,
    # remove NAs
    !is.na(latitude),
    !is.na(longitude),
    !is.na(observed_on)
  ) %>%
  mutate(
    date = ymd(observed_on),
    year = year(date),
    month = month(date, label = TRUE, abbr = FALSE),
    # Relabel entries to reduce diversity count inflation (not all species have common names)
    common_name = case_when(
      scientific_name == "Myotis" ~ "Unidentified Mouse-eared Bat",
      scientific_name == "Vespertilionidae" ~ "Unidentified Evening Bat",
      scientific_name == "Vespertilionoidea" ~ "Unidentified Vespertilionoid Bat",
      scientific_name == "Chiroptera" ~ "Unidentified Bat",
      TRUE ~ common_name # Keep all other common names as-is
    )
  )

# verify relabeling
bats_clean %>%
  filter(str_detect(common_name, "Unidentified")) %>%
  count(common_name)

# Top 10 by observation count
bats_clean %>%
  count(common_name, sort = TRUE) %>%
  print(n = 10)

# Percentage identified to species level
identified <- bats_clean %>%
  filter(!str_detect(common_name, "Unidentified")) %>%
  nrow() # 1187 unidentified

identified / nrow(bats_clean) * 100 # 64% Percentage

# Total unique observers by users
length(unique(bats_clean$user_login)) # 975

```

```

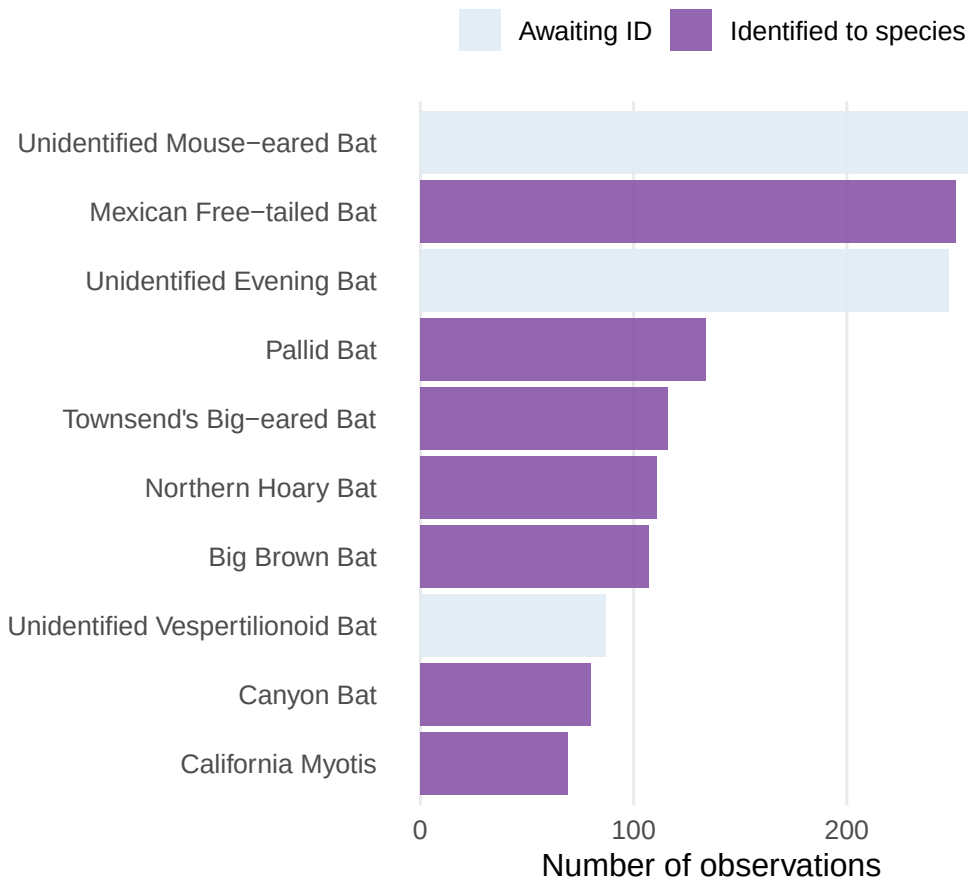
# Year range
range(bats_clean$year, na.rm = TRUE) # 1998-2025

# Plot 1
# Prepare top species data
top_species <- bats_clean %>%
  count(common_name, sort = TRUE) %>%
  slice_max(n, n = 10) %>%
  mutate(
    identified = !str_detect(common_name, "Unidentified"), # Create TRUE/FALSE column
    common_name = fct_reorder(common_name, n)
  ) %>%
ggplot(aes(x = n, y = common_name, fill = identified)) +
  geom_col(alpha = 0.9) +
  scale_fill_manual(
    values = c("FALSE" = "#E0ECF4", "TRUE" = "#8856A7"), # "#E0ECF4" "#9EBCDA" "#8856A7"
    labels = c("Awaiting ID", "Identified to species"),
    name = NULL
  ) +
  labs(
    title = "Most documented Bat Species in California",
    subtitle = "Top 10 most observed species/groups",
    x = "Number of observations",
    y = NULL,
    caption = "64% of observations identified to species level"
  ) +
  theme_minimal(base_size = 12) +
  theme(
    plot.title = element_text(size = 14, face = "bold", hjust = 0),
    plot.subtitle = element_text(size = 11, color = "gray30"),
    plot.caption = element_text(size = 9, hjust = 0, color = "gray50"),
    legend.position = "top",
    panel.grid.major.y = element_blank(), # Remove horizontal gridlines
    panel.grid.minor = element_blank(),
    axis.text = element_text(size = 10)
  )
top_species

```

Most documented Bat Species in California

Top 10 most observed species/groups



64% of observations identified to species level

```
# Plot 2

# STEP 1: Get California boundary
ca <- states() %>%
  filter(NAME == "California")

# STEP 2: Extract coordinates (for hexbin layer)
bats_coords <- bats_clean %>%
  filter(!is.na(latitude), !is.na(longitude))

# STEP 3: Create the map
plot2 <- ggplot() +
```

```

# LAYER 1: California state fill (light background)
geom_sf(data = ca, fill = "gray95", color = "gray40", linewidth = 0.5) +

# LAYER 2: Hexagonal bins showing observation density
geom_hex(
  data = bats_coords,
  aes(x = longitude, y = latitude),
  bins = 65,          # Number of hexagons (lower = bigger hexes)
  alpha = 0.8
) +

# LAYER 3: California outline on top (makes border crisp)
geom_sf(data = ca, fill = NA, color = "black", linewidth = 0.8) +
#Replace geom_hex with simple points: geom_sf(data = bats_sf, color = "#8856A7", alpha = 0.4

# COLORS: Viridis palette (colorblind-friendly)
scale_fill_viridis_c(
  option = "magma",
  name = "Raw Observations",
  direction = 1
) +

# MAP EXTENT: Zoom to California bounds
# coord_sf(xlim = c(-125, -114), ylim = c(32, 42)) +

# LABELS
labs(
  title = "Bat Observaions Across California",
  subtitle = "Hotspots of citizen science observations across the state",
  caption = "Data: iNaturalist | 1,843 observations, 1998-2025"
) +

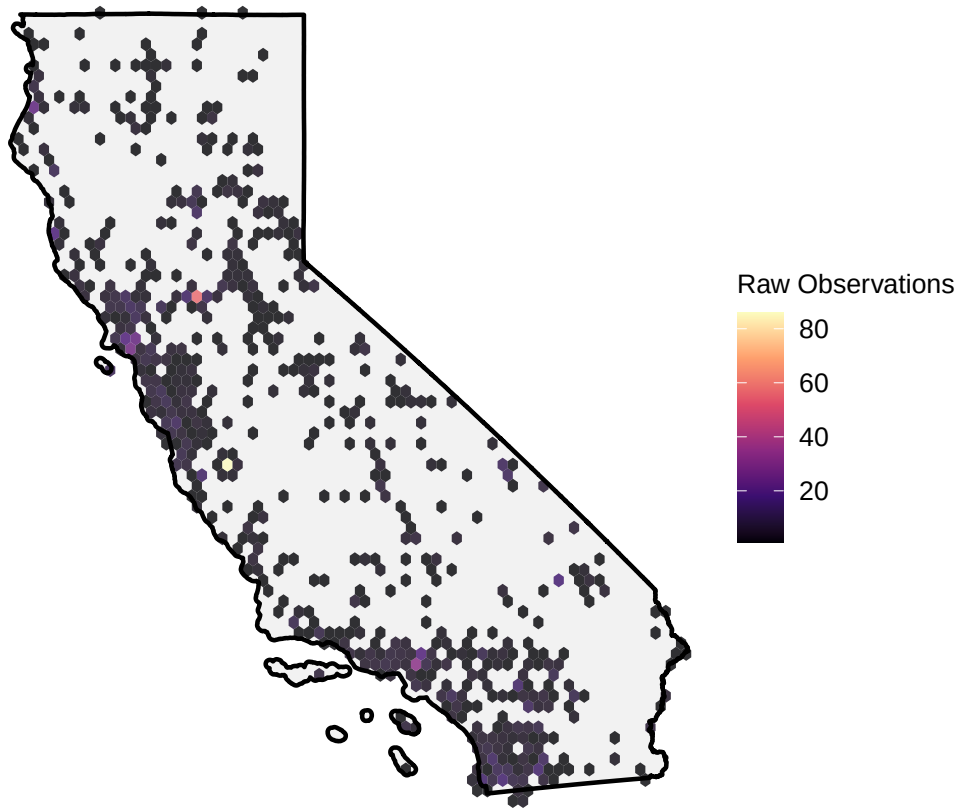
# THEME: Void removes axes, gridlines (clean for maps)
theme_void(base_size = 12) +
theme(
  plot.title = element_text(size = 14, face = "bold", margin = margin(b = 5)),
  plot.subtitle = element_text(size = 10, color = "gray30", margin = margin(b = 10)),
  plot.caption = element_text(size = 8, hjust = 0, color = "gray50"),
  legend.position = "right",
  legend.title = element_text(size = 10),
  plot.margin = margin(10, 10, 10, 10)
)

```

plot2

Bat Observaions Across California

Hotspots of citizen science observations across the state



Data: iNaturalist | 1,843 observations, 1998–2025

```
# Plot 3

# STEP 1: Prepare monthly data
monthly <- bats_clean %>%
  count(month) %>%
  mutate(month = factor(month, levels = month.name)) # Keep month order

# STEP 2: Create radial chart
plot3 <- ggplot(monthly, aes(x = month, y = n)) +
```



```

# BARS: Radial bars from center outward
geom_col(fill = "#810f7c", width = 0.9) +

# MAKE IT CIRCULAR: coord_polar converts x-axis to circle
coord_polar(start = -pi/12) + # Rotate so January starts at top

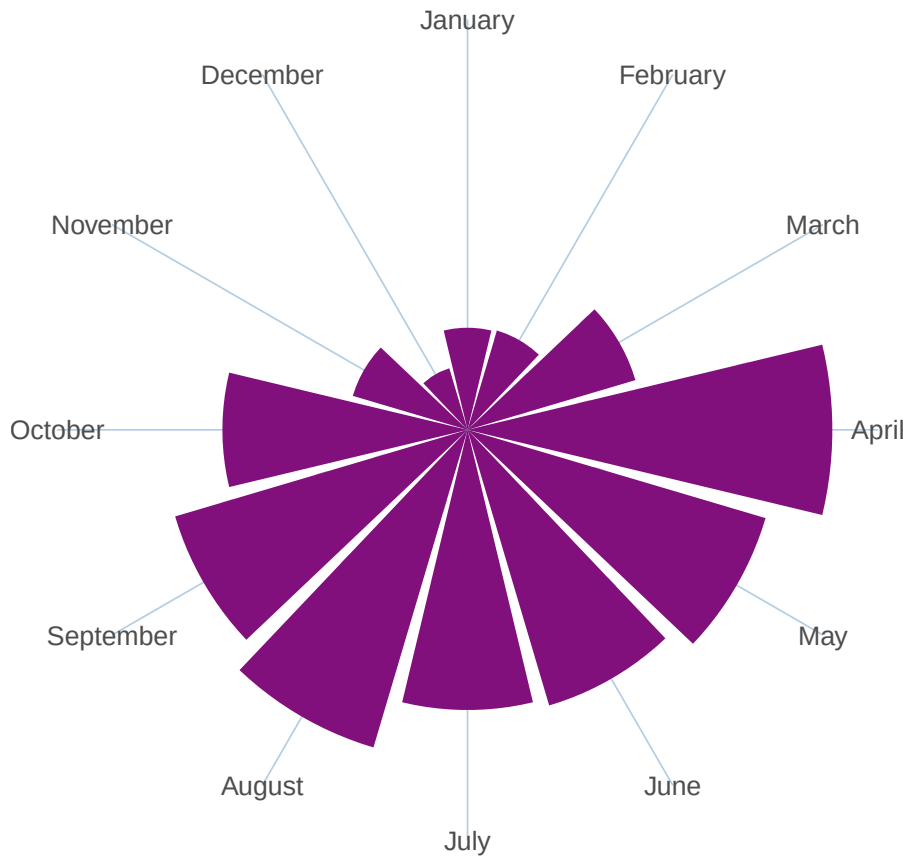
# LABELS
labs(
  title = "Seasonal observation patterns throughout the year",
  subtitle = "Peak activity in April"
) +

# THEME: Clean radial appearance
theme_minimal(base_size = 12) +
theme(
  plot.title = element_text(size = 14, face = "bold", hjust = 0.5),
  plot.subtitle = element_text(size = 10, color = "gray30", hjust = 0.5),
  plot.caption = element_text(size = 9, hjust = 0.5, color = "gray50"),
  axis.title = element_blank(), # Remove axis labels
  axis.text.y = element_blank(), # Remove radial axis numbers
  axis.ticks = element_blank(), # Remove tick marks
  panel.grid.major.x = element_line(color = "#b3cde3", linewidth = 0.3), # Month dividers
  panel.grid.major.y = element_blank(), # Remove circular gridlines
  panel.grid.minor = element_blank()
)
plot3

```

Seasonal observation patterns throughout the year

Peak activity in April



VI. Answer questions

1. What are the key insights you want your infographic to communicate, and how will your design choices help highlight and support those messages?

Three key insights I want my infographic to communicate; one California has diverse bat species documented through citizen science, two bats are found statewide across varied habitat, and three bats activity peaks seasonally. In general what information can we get from citizen science on bats in California. I hope that all my plots show citizen science efforts for example my species plot will have identified species and unidentified observation tells two stories, confirmed diversity and active citizen science participation even without expert verification. Showing community members are engaged. With my density plot I hope to show observation hotspots rather than just scattered points, revealing where citizen scientists are

most active while showing statewide distribution. I hope that my radial temporal chart emphasizes the cyclical nature of seasonal activity, making the spring/fall peaks visually intuitive and giving me a chance to be creative in my infographic design. Headers with stat numbers in bold will communicate participation and data credibility.

2. What challenges did you encounter or anticipate encountering as you continue to build / iterate on your visualizations in R? If you struggled with mocking up any of your three visualizations, describe those challenges here.

iNaturalist's `place_guess` field varies wildly from specific addresses to vague county-level descriptions. Initially tried filtering to 'urban' observations but realized text matching was capturing national parks and wildlife refuges within city names. Abandoned strict urban focus in favor of honest statewide representation. Other challenges are taxonomic information about 35% of observations aren't identified to species level (genus/family only). I decided to keep these as Unidentified categories rather than exclude them, this helps with the citizen science narrative. Density plots initially didn't display CA state outline due to coordinate system conflicts between `geom_sf()` and `stat_density_2d()`. Switch to hexbin for cleaner appearance. I'm still struggling on how to better display overall observation may need to standardize observations to percentage so it can look cleaner.

3. What ggplot extension tools / packages do you need to use to build your visualizations? Are there any that we haven't covered in class that you'll be learning how to use for your visualizations?

For now the following tools/ packages have been used.

- `sf`: to convert lat/long to spatial objects
- `tigris`: Downloaded US Census geographic boundaries for CA state outline
- `coord_polar()` : creating radial temporal chart
- `stat_density_2d()` or `geom_hex`

I haven't encounter packages not discussed in class that I might use.

4. What feedback do you need from the instructional team and / or your peers to ensure that your intended message and key insights are clear?

Is the citizen science angle clear? Does distinguishing identified vs unidentified species successfully communicate community engagement, or does it just look like messy data?

Does the density/hexbin approach work, or would simple points with high transparency be clearer for showing distribution?

Do the three plots together tell a coherent story, or do they feel disconnected? Should I add more text/annotations linking them?

Is this accessible to non-scientists, or too technical?